

Multiple Choice (10 marks)

1. Which of the following is NOT a source of atmospheric carbon?
 - (a) Respiration
 - (b) Photosynthesis
 - (c) Bacterial decomposition
 - (d) Combustion of fossil fuels
2. Which of the following depicts the correct sequence of membranes of the chloroplast, beginning with the innermost membrane and ending with the outermost membrane?
 - (a) Outer membrane, inner membrane, thylakoid membrane
 - (b) Thylakoid membrane, inner membrane, outer membrane
 - (c) Inner membrane, thylakoid membrane, outer membrane
 - (d) Thylakoid membrane, outer membrane, inner membrane
 - (e) Strama, outer membrane, inner membrane
3. Which statement about variation is true?
 - (a) All phenotypic variation is the result of genotypic variation.
 - (b) All genetic variation produces phenotypic variation.
 - (c) All nucleotide variability results in neutral variation.
 - (d) All new alleles are the result of nucleotide variability.
4. Most animal cells, regardless of species, are relatively small and about the same size. Relative to larger cells, why is this?
 - (a) Smaller cells are more resilient to external damage.
 - (b) Smaller cells have a larger number of organelles.
 - (c) Smaller cells require less energy to function.
 - (d) Smaller cells can multiply faster.
 - (e) Smaller cells fit together more tightly.
5. Short sequence by promoter that assists transcription by interacting with regulatory proteins.
 - (a) Promoter
 - (b) Inducer
 - (c) Repressor
 - (d) Operator
 - (e) Inhibitor

True / False (10 marks)

Answer True or False

6. True or False: Estrogen increases blood calcium levels, counteracting the effects of parathormone.
7. True or False: The probability of rolling a 5 on two successive rolls of a balanced die is $1/12$.
8. True or False: Pendant red-colored flowers are adaptations that limit pollination by bees and promote pollination by hummingbirds.
9. True or False: Sister chromatids separate during meiosis I and contribute to genetic diversity.
10. True or False: Blood sample analysis can effectively determine a dog's sensitivity to various sound frequencies.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the survival of bacterial cells after the removal of their cell wall by lysozyme treatment, and analyze the biochemical differences between bacterial and plant cell walls.
12. Discuss the components of plasma that influence blood pressure, analyzing their mechanisms of action, and identify other solutes present in plasma that contribute to its overall composition.

End of Paper